	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	Assignment# 2

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AGILE SOFTWARE DEVELOPMENT ASSIGNMENT

Going Agile at City General Hospital

Choosing the right method and planning the first three sprints of the Patient Care Mobile App

1. Background

City General Hospital wants a mobile app where patients can book appointments, look at their lab reports, get reminders to take their medicines, and chat with nurses. The hospital has built software before using the Waterfall model, and it did not go well. The requirements were frozen at the start, doctors and patients only saw the product at the very end, and by then the system no longer matched how the wards and clinics really work.

I am writing this as the Agile Team Lead for the new project. My job is to recommend a way of working that avoids those old mistakes, and then show how the first six weeks would actually run. The report is split into two parts. **Part A** (Task 1) explains which Agile method I picked and why. **Part B** (Task 2) is the Scrum plan with the team roles, three two-week sprints and a Definition of Done.

Part A: Agile Framework Selection

2. My recommendation: a Hybrid of Scrum and XP

After looking at the project, I recommend a **Hybrid approach**. Scrum is the main framework that organises the team and the work, and a few Extreme Programming (XP) engineering practices are added on top to keep the code safe and clean. Put simply, Scrum answers “how do we organise ourselves and listen to the hospital?” and XP answers “how do we write code that we can trust with patient data?”

Scrum gives us short sprints, a Product Owner, a Product Backlog and regular reviews (Schwaber and Sutherland, 2020). XP brings test-driven development, pair programming, continuous integration and refactoring (Beck and Andres, 2004; Agile Alliance, n.d.). Figure 1 shows how the two fit together.

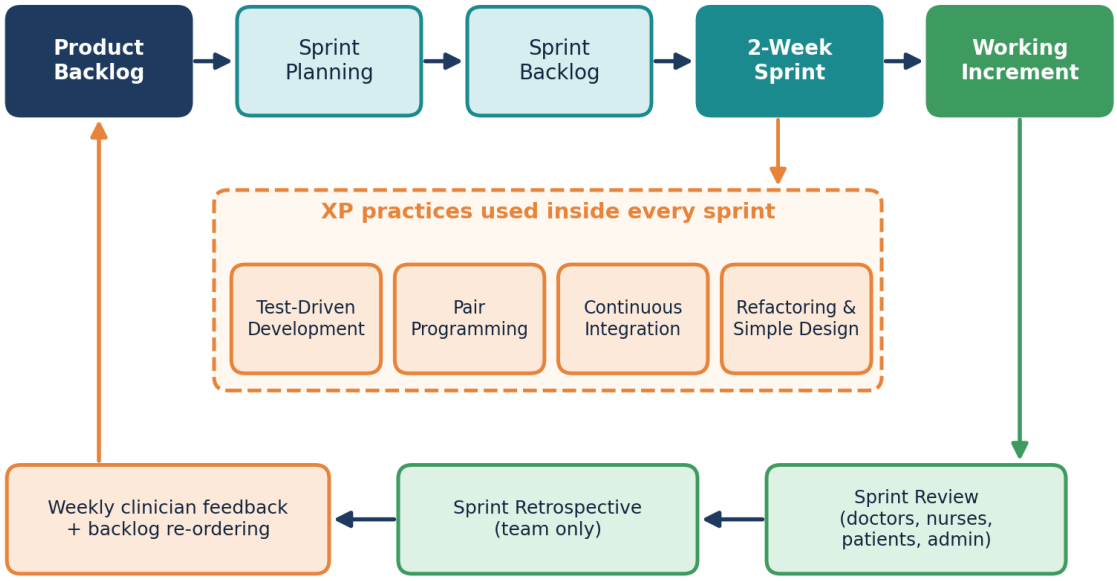


Figure 1: Scrum cycle with XP practices working inside every sprint

3. Why this choice fits the project

3.1 Team size

The Scrum Guide says a Scrum Team should be small enough to stay quick and flexible, usually ten people or fewer (Schwaber and Sutherland, 2020). Our team has nine, so it can work as one single Scrum team. We do not need to split into groups or use a heavy scaling framework. All nine people can join a 15-minute daily meeting and know what everyone is doing. Also, because the designer, QA engineers and DevOps engineer sit inside the same team, there are no hand-offs between separate departments, which was one of the quiet problems of the Waterfall projects.

3.2 Frequent feedback

Doctors, nurses, patients and the administration are free for feedback every week. A two-week sprint matches this nicely: each sprint gets a short **mid-sprint check-in** (a 30 to 45 minute demo of work in progress) and a proper **Sprint Review** at the end. That means clinicians see real software every week instead of waiting four months. If a screen does not suit a busy nurse, we find out in days, not at the end of the project.

3.3 Requirement volatility

The requirements will change as clinical staff and patients react to early versions, so we should expect change instead of fighting it. Scrum is built for this. The Product Backlog is a living list that the Product Owner can re-order at any time, and anything that has not started yet can be changed cheaply. Sommerville (2016) also points out that agile approaches suit systems where requirements

are likely to change, because plan-driven approaches make late changes expensive. Section 4 explains exactly how we handle clinicians changing their minds.

3.4 Timeline

Four months is about eight two-week sprints. The pilot in one department happens after month 2, which is after Sprint 4. Because every sprint ends with a working increment, we will already have four usable versions by the time the pilot starts, and the pilot users will not be seeing something half-finished for the first time. Fixed-length sprints also give the hospital administration a steady rhythm to follow, so nobody has to guess how far along we are.

3.5 Sensitive medical data

Scrum on its own says very little about how to write code. That is a real gap here, because lab results and medicines are among the most private things a person has. Healthcare guidance such as the HIPAA Security Rule talks about technical safeguards like access control and protecting data with encryption (U.S. Department of Health and Human Services, n.d.), and OWASP publishes security guidance for mobile apps (OWASP, n.d.). XP practices help us follow this in daily work:

- **Test-driven development (TDD):** tests for login, permissions and encryption are written before the code, so security rules are checked automatically every time.
- **Pair programming:** two people review each other live while writing the sensitive parts, such as authentication and report access.
- **Continuous integration (CI):** every code change is built and tested automatically, with a security scan, so mistakes are caught within minutes.
- **Refactoring and simple design:** keeps the code clean so it stays easy to change when clinicians ask for something new.

3.6 Why not pure Scrum or pure XP?

Pure Scrum would organise us well but leave the coding quality up to chance, which is risky with medical data. **Pure XP** works best when a customer representative is available on site almost all day. Doctors and nurses cannot sit with developers all day, they can give us weekly feedback, and XP also has fewer ready-made events for coordinating with hospital management. The Hybrid keeps the good parts of both.

Project factor	What it means for us	How the Hybrid answers it
Team size (9)	Small enough for a single team	One Scrum Team, one daily meeting, no scaling needed
Weekly feedback	Clinicians can react every week	Mid-sprint check-in plus Sprint Review every two weeks

Project factor	What it means for us	How the Hybrid answers it
Changing requirements	Priorities will move	Living Product Backlog ordered by the Product Owner
4 months, pilot at month 2	Early, real versions needed	Eight sprints, a working increment after each one
Patient data privacy	Security cannot be an afterthought	XP: TDD, pairing, CI, plus security items in the Definition of Done

4. Handling changing priorities from clinicians

This is the part where Waterfall failed, so I want to be clear about the process:

1. **One door for requests.** Every idea or complaint goes to the Product Owner and lands in the Product Backlog. Developers do not take requests directly from individual doctors, otherwise the team gets pulled in ten directions.
2. **Weekly backlog refinement.** Around the weekly feedback meeting, the Product Owner re-orders the backlog, splits big stories into smaller ones and adds acceptance criteria.
3. **Changes enter at Sprint Planning.** New or re-ordered items are picked at the start of the next sprint. During a sprint we protect the Sprint Goal, so the team can focus.
4. **Real emergencies are still possible.** If something affects patient safety, the Product Owner can discuss it with the team straight away. The Scrum Guide allows the scope to be renegotiated during a sprint as more is learned, as long as the Sprint Goal is not endangered (Schwaber and Sutherland, 2020).
5. **A small buffer.** I would keep roughly 10 to 15 percent of each sprint’s capacity free for urgent feedback fixes. This is my own planning choice, not a Scrum rule, and we can adjust it after the first retrospective.
6. **Decisions on real software.** The Sprint Review is where clinicians react to working features. Their comments become new backlog items, so the product follows their real workflow.

Part B: Scrum Implementation Plan

5. Scrum roles

Scrum has three accountabilities: the Product Owner, the Scrum Master and the Developers (Schwaber and Sutherland, 2020). Here is how I would assign our nine people, plus the one person the hospital must provide.

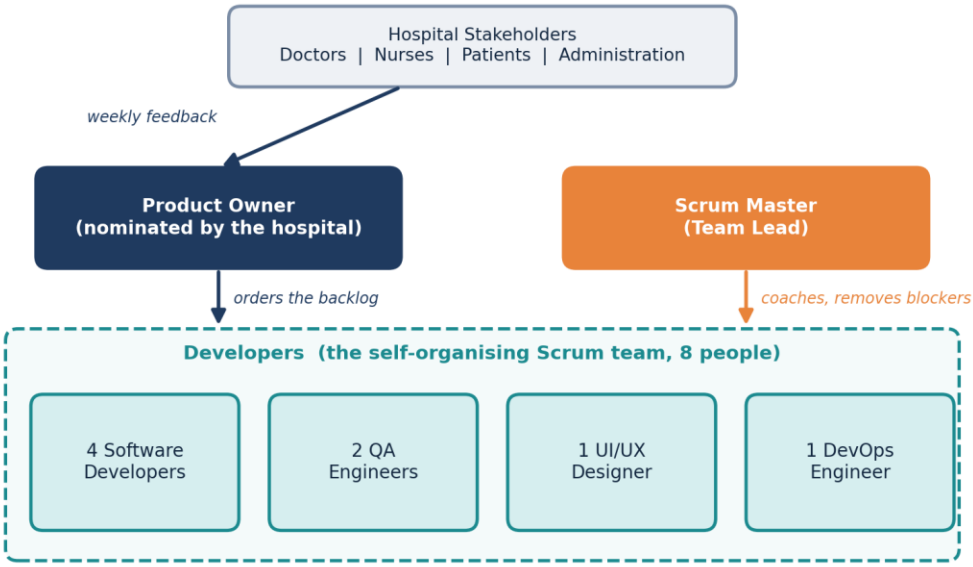


Figure 2: Scrum roles for the Patient Care App team

Role	Who	What they do
Product Owner	A senior doctor or clinical manager nominated by the hospital (not one of the nine)	Owns the Product Backlog and decides priorities. Talks to doctors, nurses, patients and administration, and accepts or rejects finished stories. The hospital must give this person real time every week, otherwise the plan breaks.
Scrum Master	Team Lead	Runs the Scrum events, removes blockers, protects the team from outside interruptions and helps everyone follow the working agreements.
Developers	4 developers	Build the app and the back end. Work in pairs on security-sensitive code and write tests first.
Developers	2 QA engineers	Join from day one: write test cases while stories are being planned, automate tests and test every story on staging. Testing is not left to the end.

Role	Who	What they do
Developers	1 UI/UX designer	Designs screens a little ahead of the developers, runs quick usability tests with nurses and patients, and checks that text sizes and layouts work for older users.
Developers	1 DevOps engineer	Builds the CI/CD pipeline, test and staging environments, handles encryption settings and secrets, and monitors the system.
Stakeholders	Doctors, nurses, patients, administration	Give weekly feedback, attend Sprint Reviews and tell us if the app fits real clinical work.

In Scrum, everyone who builds the product is called a “Developer”, whatever their skill. So the eight non-lead members form the Developers group, and the Team Lead acts as Scrum Master.

Sprint rhythm (same for every sprint)

Event	When	Time (my plan)
Sprint Planning	Day 1 of the sprint	About 4 hours
Daily Scrum	Every day, same time	15 minutes
Mid-sprint check-in with clinicians	End of week 1	30 to 45 minutes
Backlog refinement	During the sprint, with the Product Owner	About 1 hour
Sprint Review	Last day of the sprint	About 2 hours
Sprint Retrospective	After the Review, last day	About 1.5 hours

6. Release roadmap

Figure 3 puts the three sprints into the bigger four-month picture. Sprints 1 to 3 are planned in detail below. Sprints 4 to 8 are only sketched on purpose, because the backlog will change after clinicians react to the early versions.

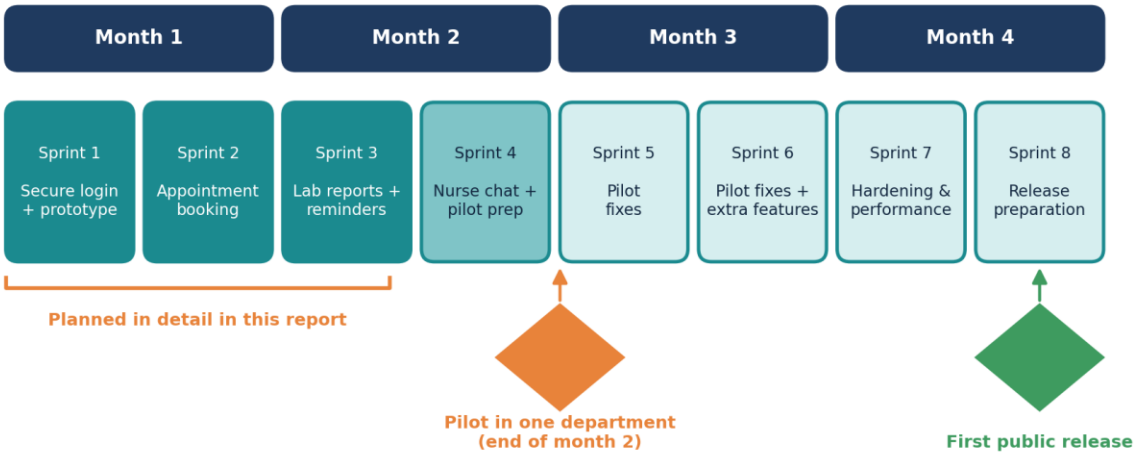


Figure 3: Four-month roadmap with the pilot and public release

7. The first three sprints

The user stories follow the usual "As a… I want… so that…" format (Cohn, 2004). Each one has acceptance criteria that the Product Owner agrees on in Sprint Planning.

Sprint 1 (Weeks 1 and 2): Secure foundation and early prototype	
Sprint Goal	Lay a secure base so that patients can create an account and log in safely, and clinicians can click through a prototype of the booking flow to tell us early if it fits their work.
Selected backlog items	• **US1:** As a patient, I want to register and log in with my mobile number and a one-time code, so that only I can open my account. • **US2:** As a patient, I want the app to log me out after a few minutes of inactivity, so that nobody can read my records if I leave my phone unlocked. • **US3:** As a hospital administrator, I want separate access rights for patients, doctors, nurses and reception staff, so that people only see data they are allowed to see. • **US4:** As a patient, I want to browse departments and doctors, so that I know who I can book with. • **US5:** As a doctor or nurse, I want to click through a prototype of the appointment flow, so that I can say early whether it matches how our clinic works. • **US6 (technical):** As the team, we want an automated CI pipeline and an encrypted test database, so that every later story is built on a secure base.
Key deliverable (increment)	A working build on the test server with registration, login, auto-logout, role-based access and the department/doctor list. The CI pipeline and encrypted database are running. A clickable booking prototype has been tried by clinicians.
Sprint Review	**Who attends:** Product Owner, whole Scrum team, 2 doctors, 1 or 2 nurses, a hospital administration representative, a hospital IT/security contact and, if possible, 2 or 3 patient volunteers. **Feedback expected:** Is login easy enough for elderly patients? Are the user roles right (for example a nurse-in-charge or receptionist)? Does the doctor list show what patients actually need? Does the prototype booking flow match the clinic's way of working?

Sprint Retrospective	**Who attends:** Scrum team only, with the Team Lead facilitating. **Focus:** How well pair programming and TDD worked in the first sprint, whether the CI pipeline was reliable, whether the stories were too big, and whether the Definition of Done needs any change.

Sprint 2 (Weeks 3 and 4): Appointment booking	
Sprint Goal	A patient can book, view, change and cancel an appointment, and a doctor can see the day's schedule.
Selected backlog items	• **US7:** As a patient, I want to choose a department, doctor and free time slot and book an appointment, so that I do not have to visit or phone the hospital. • **US8:** As a patient, I want the app to stop me from booking a slot that is already taken, so that there are no double bookings. • **US9:** As a patient, I want to see my upcoming appointments, so that I remember when to come. • **US10:** As a patient, I want to cancel or reschedule an appointment, so that the slot can be used by someone else. • **US11:** As a patient, I want a confirmation and a reminder notification, so that I do not miss my visit. • **US12:** As a doctor, I want to see my daily appointment list, so that I can prepare for the day.
Key deliverable (increment)	End-to-end appointment booking working on the staging server using test schedule data from one department, with notifications and a doctor's daily view.
Sprint Review	**Who attends:** Product Owner, Scrum team, doctors from the pilot department, 2 nurses or reception staff, administration representative and a few patient volunteers who try booking on their own phones. **Feedback expected:** Are the slot lengths realistic? How should walk-in and emergency patients be handled? How late can a patient cancel? Are the notification messages clear without showing private details?
Sprint Retrospective	**Who attends:** Scrum team only. **Focus:** How accurate our estimates were, whether QA could test stories while they were being built (not afterwards), and how well last sprint's feedback was turned into backlog items.

Sprint 3 (Weeks 5 and 6): Lab reports and medication reminders	
Sprint Goal	Patients can securely view their lab reports and receive medication reminders, ready to show the pilot department's leaders.
Selected backlog items	• **US13:** As a patient, I want a list of my lab reports with date and test name, so that I can find the one I need. • **US14:** As a patient, I want to open a report only after confirming my identity again, so that a stolen unlocked phone does not expose my results. • **US15:** As a patient, I want a notification when a new report is ready (without any result details in the message), so that I know when to check. • **US16:** As a nurse, I want to enter a patient's medication schedule (medicine, dose, times), so that the patient gets correct reminders. • **US17:** As a patient, I want reminders at the right times and a way to mark a dose as taken or snooze it, so that I stay on track. • **US18:** As an administrator, I want every access to lab reports saved in an audit log, so that we can show who viewed what and when.
Key deliverable (increment)	Lab report viewing and medication reminders working on staging, with encrypted storage and an audit log. Also a short security test report and release notes. This build is the base for the pilot, and nurse chat is planned for Sprint 4.
Sprint Review	**Who attends:** Product Owner, Scrum team, the head of the pilot department, a lab staff representative, doctors, nurses, administration representative and hospital IT/security. **Feedback expected:** Is the report layout readable for ordinary patients? Should a doctor release certain results first? Who is allowed to enter medicines? Is the reminder wording and timing practical?
Sprint Retrospective	**Who attends:** Scrum team only. **Focus:** What the security scans and QA found, a pilot readiness checklist, whether the workload for the rest of month 2 is realistic, and what we should stop, start and continue doing.

8. Definition of Done

A story is only "Done" when it meets every point below. A shared Definition of Done means everyone understands the same quality level (Agile Alliance, n.d.), and for a hospital app I have added security and privacy checks to the normal quality ones. Figure 4 shows the path a story follows.

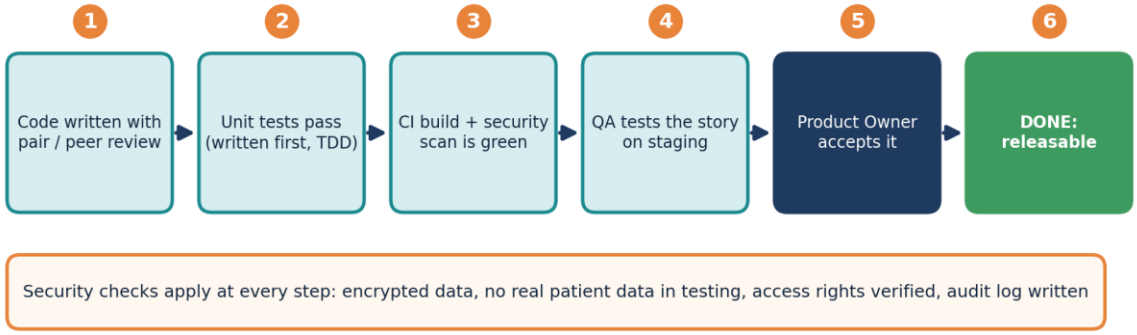


Figure 4: Path of a user story to "Done"

Code and testing

- Code is written in a pair or reviewed by a second developer before it is merged.
- Unit tests were written first (TDD) and all automated tests pass in the CI pipeline.
- New code is covered by automated tests (our team target is at least 80 percent).
- QA has run the test cases for the story on the staging server and there are no open critical or high-priority bugs.

Security and privacy

- Data is encrypted in transit (TLS) and at rest, and no passwords, tokens or keys are stored in plain text.
- The automated dependency and security scan shows no high or critical issues.
- Access rights for the story are tested, so a patient cannot see another patient's data and staff only see what their role allows.
- Only dummy test data is used in development and testing, never real patient records.
- Notifications and logs contain no private medical details, and access to sensitive records is written to the audit log (see OWASP, n.d., and U.S. Department of Health and Human Services, n.d.).

Usability and acceptance

- The UI/UX designer has checked the screens against the prototype, including readable text sizes for older users.

- The Product Owner has checked the acceptance criteria and accepted the story.
- Technical notes and user notes are updated, and the feature is deployed through the automated pipeline, not by hand.

9. Conclusion

Waterfall failed at City General Hospital because the hospital saw the product too late and could not change its mind. A Hybrid of Scrum and XP fixes both problems: Scrum puts working software in front of clinicians every two weeks and lets the Product Owner re-order priorities, while XP practices and a strict Definition of Done protect patient data while we move fast. With a single nine-person team, weekly stakeholder feedback and a pilot after Sprint 4, this combination fits the project well. The plan is not meant to be fixed; after each Sprint Review and Retrospective we will adjust it, and that is exactly the point of working in an Agile way.

References

Agile Alliance (n.d.) Definition of Done. Available at: https://www.agilealliance.org/glossary/definition-of-done/

Agile Alliance (n.d.) Extreme Programming (XP). Available at: https://www.agilealliance.org/glossary/xp/

Atlassian (n.d.) Scrum: what it is, how it works, and why it's awesome. Available at: https://www.atlassian.com/agile/scrum

Beck, K. and Andres, C. (2004) Extreme Programming Explained: Embrace Change. 2nd edn. Boston: Addison-Wesley.

Cohn, M. (2004) User Stories Applied: For Agile Software Development. Boston: Addison-Wesley.

OWASP (n.d.) OWASP Mobile Application Security. Available at: https://owasp.org/www-project-mobile-app-security/

Schwaber, K. and Sutherland, J. (2020) The Scrum Guide. Available at: https://scrumguides.org/scrum-guide.html

Sommerville, I. (2016) Software Engineering. 10th edn. Harlow: Pearson.

U.S. Department of Health and Human Services (n.d.) The HIPAA Security Rule. Available at: https://www.hhs.gov